

Theory

Gaussian Beam Optics

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Abstract

1 Experiment specifics

1. Wavelength: 1064nm
2. Collimator output beam size: $0.56\text{ mm } \frac{1}{e^2}$ diameter
3. PPLN crystal: $50 \times 2 \times 0.5\text{ mm}^3$
4. Previous best measured output: 57 mW green, PPLN at 59.5°C
5. Previous calculated waist: about $45\text{ }\mu\text{m}$
6. Desired next target: about $77\text{--}80\text{ }\mu\text{m}$

2 Definitions and Equations

Beam Diameter ($\frac{1}{e^2}$): $\frac{1}{e^2} = 0.135335$ beam diameter captures the diameter where the intensity of the beam drops to 13.53% or in other words 86.47% of the beam intensity is within the given diameter.

figures/Picture1en-us.jpg

Beam Waist (ω_0): The beam waist is the portion of the beam where the radius is minimized. This is notated as w_0 . The tighter the focus the smaller the waist, but this introduces a large amount of divergence. This can be seen in the z dependent beam width equation

$$\omega^2(z) = \omega_0^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_0^2} \right)^2 \right] \quad (1)$$

Small waists are achieved by a "short focal length and a high numerical aperture largely filled by the input beam." [?]

figures/download.png

Radius of Curvature (R_g): The radius of curvature defined by Eq. ???. At its boundaries, $\lim_{z \rightarrow 0} R_g(z) = \infty$ where the wave front is perfectly vertical and acts like a section of a circle with an infinite radius. $\lim_{z \rightarrow \infty} R_g(z) = z$ where the wave front averages out to a radius equal to the distance from the waist.

$$R_g(z) = z \left[1 + \left(\frac{\pi \omega_0^2}{\lambda z} \right)^2 \right] \quad (2)$$

Rayleigh Range (z_0): Referring back to R_g , the Rayleigh range is the distance where the cross sectional area of the beam is no more than twice the cross sectional area of the beam waist. This ensures we maintain at least 50% of the beam. [?]

$$z_0 = \frac{\pi \omega_0^2}{\lambda} \quad (3)$$

Beam Divergence (θ): Is the slope of the linear component of the beam expansion, also known as the expansion in the far field. This is measure with half angles dependent on the wavelength and inverse of the waist defined by Eq. ???.

$$\theta = \frac{\lambda}{\pi \omega_0} \quad (4)$$

Screenshot 2026-06-02 at 13.18.55.png

Confocal Parameter (b): A use of the Rayleigh range is the Confocal parameter, the distance or error in which the focus of your beam can operate at a level of usability. It is the "the total length of the focused region around the beam waist" [?].

Boyd-Kleinman Focusing: $h = 1.068$ at $L/b = 2.84$ [?] $h = 1.066$ at $L/b = 2.81$ [?] Where h is conversion efficiency, L is this length of the crystal and b is the confocal parameter which is twice the Rayleigh range. So we should work backwards to find the waist. Refer to calculation 2

1

2

3 Calculations

1. Starting from a $0.56 \text{ mm } 1/e^2$ beam diameter at 1064 nm , what are the approximate Gaussian beam parameters?

Using Eq. ??³

$$z_0 = \frac{\pi \omega_0^2}{\lambda} = \frac{\pi \cdot (0.00028)^2}{0.000001064} = 0.23148577447 \text{ m} \approx 23.148 \text{ cm}$$

Using Eq. ??

$$\theta = \frac{\lambda}{\pi \omega_0} = \frac{0.000001064}{\pi \cdot 0.00028} = 0.001209578 \approx 1.210 \times 10^{-3}$$

2. (a) Given $\frac{L}{b} = 2.84$ and $L = 50 \text{ mm}$ [?]

$$b = \frac{L}{2.84} = \frac{0.05}{2.84}$$

Using Eq. ??

$$\frac{0.05}{2.84} = b = 2 \cdot z_0 = \frac{2 \cdot \pi \cdot \omega_0^2}{\lambda}$$

$$\begin{aligned} \frac{0.05}{2.84} &= \frac{2 \cdot \pi \cdot \omega_0^2}{1064 \cdot 10^{-9}} = \sqrt{\frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot 2.84}} = \omega_0 \\ &\approx 0.000054601768829654717265993916359 \approx 54.602 \text{ } \mu\text{m} \end{aligned}$$

- (b) Given $\frac{L}{b} = 2.81$ and $L = 50 \text{ mm}$ [?]

$$b = \frac{L}{2.81} = \frac{0.05}{2.81}$$

Using Eq. ??

$$\frac{0.05}{2.81} = b = 2 \cdot z_0 = \frac{2 \cdot \pi \cdot \omega_0^2}{\lambda}$$

$$\begin{aligned} \frac{0.05}{2.81} &= \frac{2 \cdot \pi \cdot \omega_0^2}{1064 \cdot 10^{-9}} = \sqrt{\frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot 2.81}} = \omega_0 \\ &\approx 0.00005489246353086303938501371680382 \approx 54.892 \text{ } \mu\text{m} \end{aligned}$$

(c) Given $\frac{L}{b} = 1$ and $L = 50 \text{ mm}$ [?]⁴

$$b = \frac{L}{1} = 0.05$$

Using Eq. ??

$$0.05 = b = 2 \cdot z_0 = \frac{2 \cdot \pi \cdot \omega_0^2}{\lambda}$$

$$\begin{aligned} 0.05 &= \frac{2 \cdot \pi \cdot \omega_0^2}{1064 \cdot 10^{-9}} = \sqrt{\frac{1064 \cdot 10^{-9}}{2 \cdot \pi}} = \omega_0 \\ &\approx 0.0000920165364078046129480467 \approx 92.017 \text{ } \mu\text{m} \end{aligned}$$

(d) If we want to hit the previous calculated waists and predicted targets lets work backwards, letting $\frac{L}{b} = \alpha$

i. $45 \mu\text{m} \Rightarrow$

$$\begin{aligned} \omega_0 &= 0.000045 = \sqrt{\frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha}} \\ 0.000045^2 &= \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha} \Rightarrow \alpha = \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot 2.025 \times 10^{-9}} \approx 4.181 \end{aligned}$$

Too narrow of a waist which leads to too much diffraction, and loses efficiency.

ii. $77 \mu\text{m} \Rightarrow$

$$\begin{aligned} \omega_0 &= 0.000077 = \sqrt{\frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha}} \\ 0.000077^2 &= \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha} \Rightarrow \alpha = \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot 5.929 \times 10^{-9}} \approx 1.428 \end{aligned}$$

In the proper range

iii. $80 \mu\text{m} \Rightarrow$

$$\begin{aligned} \omega_0 &= 0.00008 = \sqrt{\frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha}} \\ 0.00008^2 &= \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot \alpha} \Rightarrow \alpha = \frac{1064 \cdot 10^{-9} \cdot 0.05}{2 \cdot \pi \cdot 6.4 \times 10^{-9}} \approx 1.323 \end{aligned}$$

In the proper range

3. Now trying without the assumption of the waist being at the front face of the collimator⁵

(a) Using Eq. ??

$$\begin{aligned} \theta &= \frac{\lambda}{\pi \omega_{01}} \Rightarrow \omega_{01} = \frac{\lambda}{\pi \theta} = \frac{0.000001064}{\pi \cdot 0.00121} \\ &\approx 0.000279902247024424 \approx 0.2799 \text{ mm} = \omega_{01} \end{aligned}$$

This is too thick to aim for as the waist so we will have to magnify the beam

(b) Now using Eq. ?? from Eq. 2-27 [?]

$$\omega^2(z) = \omega_{01}^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \right] \quad (5)$$

$$\begin{aligned} \frac{\omega^2(z)}{\omega_{01}^2} &= 1 + \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \Rightarrow \left(\frac{\omega^2(z)}{\omega_{01}^2} - 1 \right) = \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \\ \Rightarrow \left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1} \right) \pi \omega_{01}^2 &= \pm \lambda z \Rightarrow \left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1} \right) \frac{\pi \omega_{01}^2}{\lambda} = \pm z \end{aligned}$$

Using $\omega(z) = 0.00028$ m as diameter of the collimator is 0.56 mm

$$\Rightarrow \left(\sqrt{\frac{0.00028^2}{0.0002799^2} - 1} \right) \cdot \frac{\pi \cdot 0.0002799^2}{0.000001064} = \pm z \approx \pm 0.006184 = \pm 6.184 \text{ mm}$$

Which is understood to be the distance in the negative z direction that the waist is located behind the collimator face.

(c) *Now to recalculate with the new beam waist location* Let $w_0 = 0.00052$ –

(d) Starting with

$$q_{out} = \frac{A(iz_{01}) + B}{C(iz_{01}) + D}$$

we have a given $q_{in} = 0.0002799$ m and a desired $q_{out} = 0.00008$ m Separating the complex components results in

$$q_{out} = \left[\frac{BD + ACz_{01}^2}{D^2 + C^2z_{01}^2} \right] + i \left[\frac{(AD - BC)z_{01}}{D^2 + C^2z_{01}^2} \right]$$

(e) *Calculating the required lens and crystal placement*

To achieve our target waist of 80 μm using our standard $f = 88.3$ mm lens, we must first determine the required magnification M :

$$M = \frac{\omega_{02}}{\omega_{01}} = \frac{0.080 \text{ mm}}{0.2799 \text{ mm}} \approx 0.2858$$

Using the Gaussian magnification factor derived from the ABCD matrix transformations defined by Osche [4], we can solve for the required distance from the true beam waist to the lens (d_1):

$$M = \frac{f}{\sqrt{(d_1 - f)^2 + z_{01}^2}} \Rightarrow d_1 = f + \sqrt{\left(\frac{f}{M} \right)^2 - z_{01}^2}$$

4 Use case

PPLN: To get the most out of our PPLN crystals, there are four key aspects that you need to consider:⁶

1. Crystal Length

(a) Don't worry about it

2. Polarization

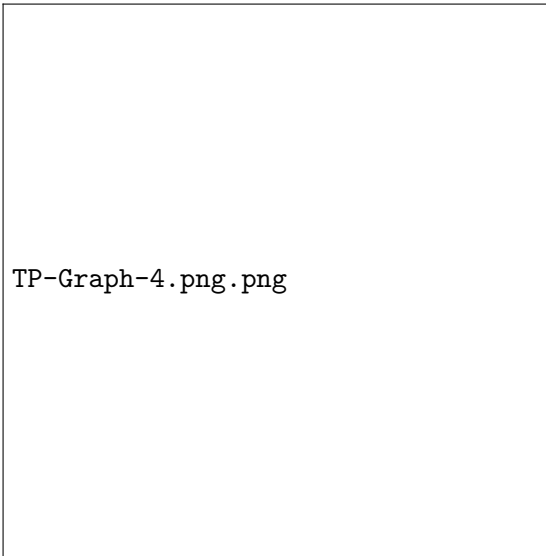
- (a) The input light must e-polarized, i.e. the polarization must be aligned with the dipole moment of the crystal. This is accomplished by aligning the polarization axis of the light parallel to the thickness of the crystal. [?]

3. Focusing and the optical arrangement

- (a) For SHG with CW lasers, a theoretical result from Boyd and Kleinmann shows that optimum efficiency can be achieved when the ratio of the crystal length to the confocal parameter is 2.84. [?]

4. Temperature and period

- (a) The crystal is manufactured to have "stripes" of magnetically opposite layers with a distance in between labeled as the poling period. IT in theory helps create quasi-phase-matched light so the power continually builds up.
- (b) Because of manufacturing inconsistencies and because heat transfer allows the crystal to expand or contract, we can change the poling period of the crystal by changing the crystals temperature.
- (c) Note: The refractive index of the crystal also changes with temperature.
- (d) The intensity of light follows a $\text{sinc}(x) = \frac{\sin^2(x)}{x^2}$ curve as shown below, describing a crystal temperature acceptance bandwidth.



- (e) Note: Acceptance bandwidth is inversely proportional to crystal length.
- (f) The optimum temperature can be determined by heating the crystal to 20°C higher than the calculated temperature and then allowing the crystal to cool whilst monitoring the output power at the generated wavelength.[?]



Figure 1: S1 $\rightarrow$ Laser at 1064 nm

Notes

¹Wikipedia entry: Second-harmonic generation (SHG), also known as frequency doubling, is the lowest-order wave-wave nonlinear interaction that occurs in various systems, including optical, radio, atmospheric, and magnetohydrodynamic systems. As a prototype behavior of waves, SHG is widely used, for example, in doubling laser frequencies. SHG was initially discovered as a nonlinear optical process in which two photons with the same frequency interact with a nonlinear material, are "combined", and generate a new photon with twice the energy of the initial photons (equivalently, twice the frequency and half the wavelength), that conserves the coherence of the excitation. It is a special case of two-photon sum-frequency generation, and more generally of harmonic generation.

²**Phase-matching:** For sum-frequency generation to occur efficiently, phase-matching conditions must be satisfied: $\hbar k_3 \approx \hbar k_1 + \hbar k_2$ where k_1, k_2, k_3 are the angular wavenumbers of the three waves as they travel through the medium. (Note that the equation resembles the equation for conservation of momentum.) As this condition is satisfied more and more accurately, the sum-frequency generation becomes more and more efficient.

³In this calculation I assume $\frac{w_0}{2} = 0.56$ mm because it came out of a collimator. That being said this is making an assumption that the output beam may have its beam waist a mild focus somewhat away from the face of the collimator as outlined in [?] "The purpose of a beam collimator is essentially to transform a strongly diverging light beam into a collimated beam, i.e., a beam where light propagates essentially only in one direction, and the beam divergence is weak. The output beam may have its beam waist close to the output aperture, or a mild focus somewhat away from it."

⁴"In general, a good rule of thumb is that the spot size should be chosen such that the Rayleigh range is half the length of the crystal." If $z_0 = L/2$, then $b = 2z_0 = L \Rightarrow \frac{L}{b} = 1$

⁵Note: 1.21 mRad was a measured value, substantially lower than the spec sheet citing roughly 5 mRad

⁶Periodically poled lithium niobate (PPLN) is a lithium niobate crystal that switches polarity in a periodic fashion and is used for achieving quasi-phase-matching in nonlinear optics.